INCH-POUND
A-A-52510B
February 16, 2010
SUPERSEDING
A-A-52510A
09 June 2009

COMMERCIAL ITEM DESCRIPTION

GENERATOR, TACHOMETER

The General Services Administration has authorized the use of this commercial item description (CID) for all Federal Agencies.

1. SCOPE

This CID covers requirements for a direct current (DC) generator, tachometer which is used to measure rotational velocity/speed by converting mechanical rotation into a DC output voltage. This item is used on, but is not limited to, the M901 TOW anti-tank combat vehicle.

2. SALIENT CHARACTERISTICS

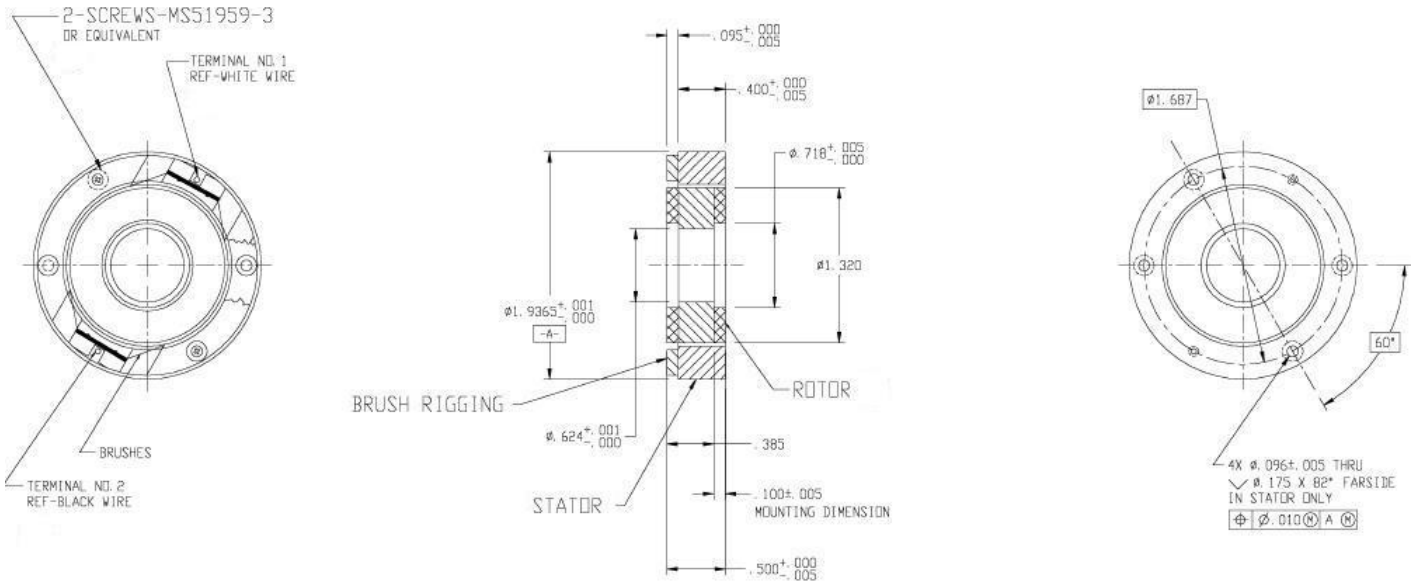
2.1 Materials. The materials used in the manufacture of the generator, tachometer shall be in accordance with the manufacturer's materials specifications as required to meet the salient characteristics of this CID. The use of recovered material made in compliance with regulatory requirements is acceptable providing that all requirements of this CID are met (see 5.4).

2.2 Design and construction. Unless otherwise specified in figure 1, the design and construction of the generator, tachometer shall be in accordance with the manufacturer's specifications and drawings. The major components of the generator, tachometer are: rotor, stator (field) with keeper, and brush assembly. These components when supplied individually for assembly of the generator, tachometer at installation, shall come only in matched sets (see 2.6.2).

Beneficial comments, recommendations, additions, deletions, clarifications, etc. and any data that may improve this document should be sent to DAMI_STANDARDIZATION@conus.army.mil or U.S. Army RDECOM, Tank Automotive Research, Development and Engineering Center, ATTN: RDTA-EN/STND/TRANS MS #268, 6501 E. 11 Mile Road, Warren, MI 48397-5000. Since contact information can change, you may want to verify the currency of this address information using the ASSIST Online database at https://assist.daps.dla.mil.

AMSC N/A FSC 6680

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PIN	Former Part Number
A52510-1	12265503

NOTES:

1. Each terminal (1 and 2) to accept one 20 AWG stranded wire maximum. Brush rigging terminal marking (1 and 2) to be installed to match stator terminal marking (1 and 2).
2. Dimensions are in inches. Unless otherwise specified, tolerances are ± 0.005.

FIGURE 1. Generator, tachometer.

2.3 Technical characteristics. The technical characteristics of the generator, tachometer shall be as specified in table I.

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TABLE I. Technical characteristics.

Technical characteristics	Value
DC resistance	87.3 ohms ± 1.25 ohms
Insulation resistance	> 100 megohms at -65°F (-54°C) to 77°F (25°C) >25 megaohms at 140°F (60°C) with 100V DC applied
Dielectric withstanding voltage	200V rms at 60 Hz for 60 seconds with less than 1mA leakage current
Speed range (normal)	20.94 to 209.40 milliradians per second
Generator (voltage) sensitivity	0.194 ± 10% volts per radian per second
Load resistance (output)	10 000 ohms minimum
Ripple voltage	± 7% average to peak

2.3.1 Verification of technical characteristics. The technical characteristics shall be verified in accordance with the following tests. Normal room conditions are defined as a temperature of 77 ± 18 °F (25 ± 10 °C), relative humidity of 20 to 80 percent, and normal atmospheric pressure.

a. DC resistance. The assembled generator shall have direct current resistance of 87.3 ± 1.25 ohms under normal room conditions.
b. Insulation resistance. The assembled generator shall have an insulation resistance of greater than 100 megohms at -65.2°F to 77°F and greater than 25 megohms at 140°F with 100 Vdc applied between the generator terminals and the rotor stack.
c. Dielectric withstanding voltage. The dielectric withstanding voltage of the electrical insulation shall be 200 Vrms, at 60 Hz applied for 60 seconds between the generator terminals and the rotor stack under normal room conditions.
d. Generator sensitivity. Generator sensitivity in either direction in the normal speed range (20.94 milliradian per second to 209.40 milliradian per second) shall be 0.194 ± 0.0194 V per radian per second. The tolerance shall include all effects of nonlinearity. The generator sensitivity, using an output load of 10,000 ohms, shall be calculated from the following formula:

$$\text{Sensitivity (volt/radian/second)} = \frac{\text{Volts X Seconds}}{\text{Revolutions X 6.28}}$$

Nonlinearity tests shall be performed under normal room conditions with speeds selected between 6 and 600 milliradians per second. Two tests shall be performed with speeds at least 500 milliradians per second apart.

e. Ripple. The generator percent ripple, using an output load of 10,000 ohms, shall be calculated from the following formula:

$$\text{Percent ripple} = \frac{\text{100 X difference voltage}}{\text{average signal level voltage}}$$

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Tests shall be performed under normal room conditions with speeds selected between 6 and 600 milliradians per second. Two tests shall be performed with speeds at least 500 milliradians per second apart.

2.4 Environmental. The generator, tachometer shall be capable of performing during and after exposure to the environmental conditions specified in table II without damage or deterioration to the technical characteristics specified in table I.

TABLE II. Environmental conditions.

Environmental condition	Mode of operation	Parameter
Temperature	(a) operating (b) nonoperating	-26°F to 158°F -65°F
Relative humidity (RH)	(a) operating (b) nonoperating	100% RH at 85°F to not > 5% RH at 125°F for 24 hours. 100% RH at 80°F to not > 2% RH at 155°F for 24 hours.
Atmospheric pressure (AP)	(a) operating (b) nonoperating	AP equivalent to an altitude of 8000 feet (min) for 6 hours. AP equivalent to an altitude of 40 000 feet (min) for 1 hour.
Shock	nonoperating	25 g half-sine wave shock impulse for 11 milliseconds in each of three mutually perpendicular axes.
Vibration	nonoperating	Simple harmonic motion applied along three mutually perpendicular axes, peak acceleration of 10 g at all frequencies, frequency range 20 to 2000 Hz, frequency sweep varied logarithmically (min. to max. only), sweep time 35 ± 5 minutes, vibrate for 5 minutes at each observed resonant frequency.

2.5 Electromagnetic interference. The generator, tachometer shall comply with emissions and Susceptibility requirements of MIL-STD-461, class A3, equipment when operated at all speeds from 20.94 to 209.4 milliradians per second with an output load of 10,000 ohms (minimum). This shall be verified in accordance with MIL-STD-461, class A3 equipment.

2.6 Identification and marking. Identification and marking of the generator, tachometer shall be permanent and legible and shall include, as a minimum, the part or identification number (PIN) "A52510-1" and the manufacturer's CAGE code and part number (see 5.2).

2.6.1 Warning label. A red warning label shall be applied to the stator (field) containing the following warning in large, bolt print, "CAUTION - DO NOT REMOVE KEEPER UNTIL ROTOR IS FULLY IN PLACE".

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2.6.2. Matched sets. In addition to other markings, tachometer generators that come in matched sets to complete the assembly shall be appropriately marked and identified as such.

3. QUALITY ASSURANCE PROVISIONS

3.1 Responsibility for inspection. The contractor is responsible for the performance of all inspections (examinations and tests).

3.2 Contractor certification. The contractor shall certify and maintain substantiating evidence that the product offered meets the salient characteristics of this CID and that the product conforms to the producer's own drawings, specifications, workmanship standards, and quality assurance practices. Items with known defects shall not be submitted for Government acceptance. The Government reserves the right to require proof of such conformance prior to the first delivery and thereafter as may be otherwise provided for under the provisions of the contract.

4. PRESERVATION, PACKAGING, PACKING, UTILIZATION, AND MARKING.

Unless otherwise specified in the contract or order, commercial packaging in accordance with ASTM D3951 shall be the minimum requirements for packaging of the generator, tachometer or any of the major components which are packaged as a matched set with a quantity per unit package of one each (see 5.2).

5. NOTES.

(This section contains information of a general or explanatory nature that may be helpful but is not mandatory.)

5.1 Addresses for obtaining copies of referenced documents.

5.1.1 Government documents. Copies of MIL-STD-461 "Electromagnetic Emission and Susceptibility Requirements for the Control of Electromagnetic Interference" are available from the Defense Printing Service Detachment Office, Bldg. 4D (Customer Service), 700 Robbins Avenue, Philadelphia, PA 19111-5094.

5.1.2 Non-Government publications. Copies of ASTM D3951 "Standard Practice for Commercial Packaging" are available from the American Society for Testing and Materials, 100 Barr Harbor Drive, West Conshohocken, PA 19428-2959.

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5.2 Ordering data. Acquisition documents must specify the following:

a. Title, number, and date of this CID.
b. If required, the specific issue of the individual documents referenced.
c. Generator, tachometer PIN number and quantity required.
d. Applicable level and packaging requirements if other than as specified.

5.3 Cross-reference data. Tachometer generators conforming to this CID are interchangeable with tachometer generators conforming to MIL-G-62363A(AT), dated 18 December 1985.

5.4 Regulatory requirements. The offeror/contractor is encouraged to use recovered materials in accordance with Public Law 94-580 to the maximum extent practicable.

MILITARY INTEREST:

Custodian:
Army - AT

CIVIL AGENCY COORDINATING ACTIVITY:
GSA - FSS

Preparing Activity:
Army - AT

(Project 6680-2010-001)

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